Tarun Singh

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Education

Bachelor of Technology in Computer Science (Data Science) **Expected Graduation: 2027**
Institute of Engineering Science, IPS Academy Indore **CGPA: 8.3**

Technical Skills

Languages: Python, SQL
Machine Learning: Scikit-learn, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning
Deep Learning: TensorFlow, Keras, PyTorch, CNN, RNN, LSTM, Autoencoders, Transformers, OpenCV
Generative AI: LLM Integration, Prompt Engineering, RAG, LangChain, LangGraph, Hugging Face, Vector Databases
Data Science: Pandas, NumPy, Matplotlib, Data Cleaning, EDA, Statistical Analysis
MLOps: FastAPI, API Integration, Git, GitHub, MLflow, Model Deployment, Docker(Basics)
Databases: PostgreSQL, MySQL, SQLAlchemy, Query Optimization
Tools: VS Code, Jupyter Notebook, Antigravity, Claude Code, GitHub Copilot

Experience

Software Engineer Trainee & Quantum Engineer ***Aug 2026 – Present***
AlphaInfo Lab

- Undergoing structured training in **Python programming**, mathematical foundations, and computational concepts relevant to **Quantum Computing**.
- Studying **Linear Algebra** with emphasis on **vectors, matrices, vector operations, and mathematical representations** used in quantum computing.
- Developing an understanding of **vectors and their applications in Quantum Algorithms**, including their role in representing and manipulating quantum states.
- Using Python and NumPy to implement linear algebra concepts includes vector operations, matrix multiplication, norms, etc.

Data Science Intern ***May 2026 – July 2026***
Alfido Tech

- Worked on a **Zomato restaurant data science project** involving **data cleaning, preprocessing, exploratory data analysis, and feature engineering**.
- Analyzed a structured dataset containing **56,252 restaurant records and 13 features** using **Python, Pandas, NumPy, Matplotlib, and Seaborn**.
- Performed **data preprocessing** including handling missing values, transforming categorical features, and preparing datasets for **statistical analysis and predictive modeling**.
- Gained practical exposure to **end-to-end data science workflows**, deployment concepts, and AI/ML-oriented services.

Projects

AI Powered Intrusion Detection System (IDS/IPS)

- Developed an AI-driven cybersecurity platform for intrusion detection using **Machine Learning and Deep Learning.**
- Built an end-to-end ML pipeline covering **data preprocessing, feature engineering, model training, evaluation, and anomaly detection**.
- Integrated **MLP and Autoencoder-based approaches** for detecting anomalous network behavior.
- Designed deployment-ready architecture incorporating **monitoring, explainable AI, and AI agent concepts**.

RAG-Intelligence

- Developed a **Retrieval-Augmented Generation (RAG)** application for intelligent document querying using **FastAPI, LangChain, Pinecone, and OpenRouter**.
- Built a document ingestion pipeline supporting **PDF, DOCX, PPTX, CSV, XLSX, TXT, Markdown, and image files**, with document extraction, chunking, metadata handling, and vector indexing.
- Implemented **semantic similarity retrieval** with **Pinecone vector database** and Google Generative AI embeddings.
- Integrated an **LLM-based RAG chain** with document-level filtering, returning answers along with relevant information.
- Created a **Streamlit frontend** for document upload, indexing, document management, and conversational interaction.

Certifications

- Completed **Python and Data Science training** at Universal Informatics, Indore.
- Earned **Python and SQL certification from IIT Bombay** through the Spoken Tutorial program.

Achievements

- Co-author of Hindi poetry anthology **Abhivyakti** and **Vice President** of the Poetry Club.
- Maintained a **1000+ days Duolingo streak**; proficient in English and currently learning Japanese.